

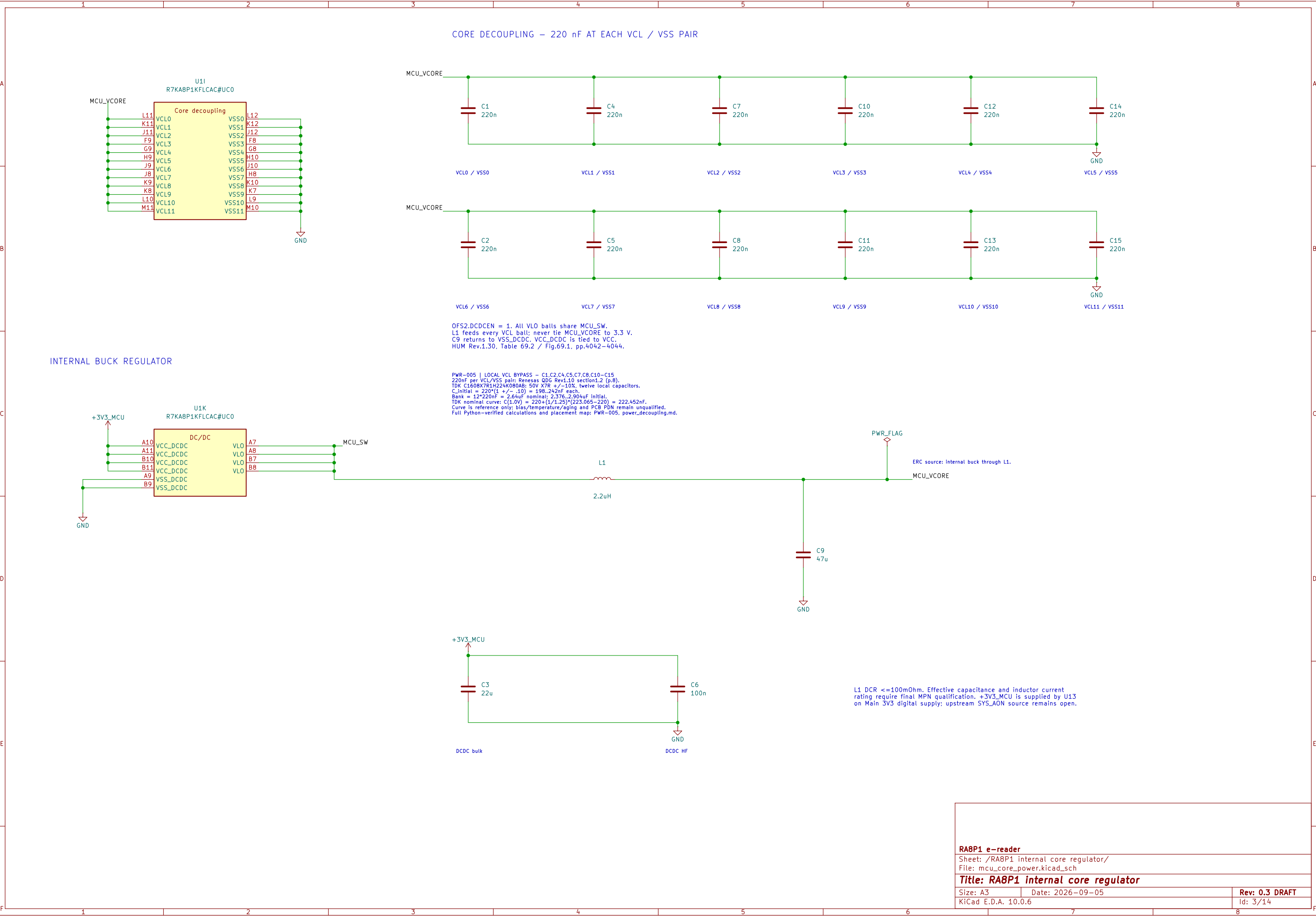
DRAFT: Existing processor units retained. Signal allocation and clocks/debug wiring are not yet complete.

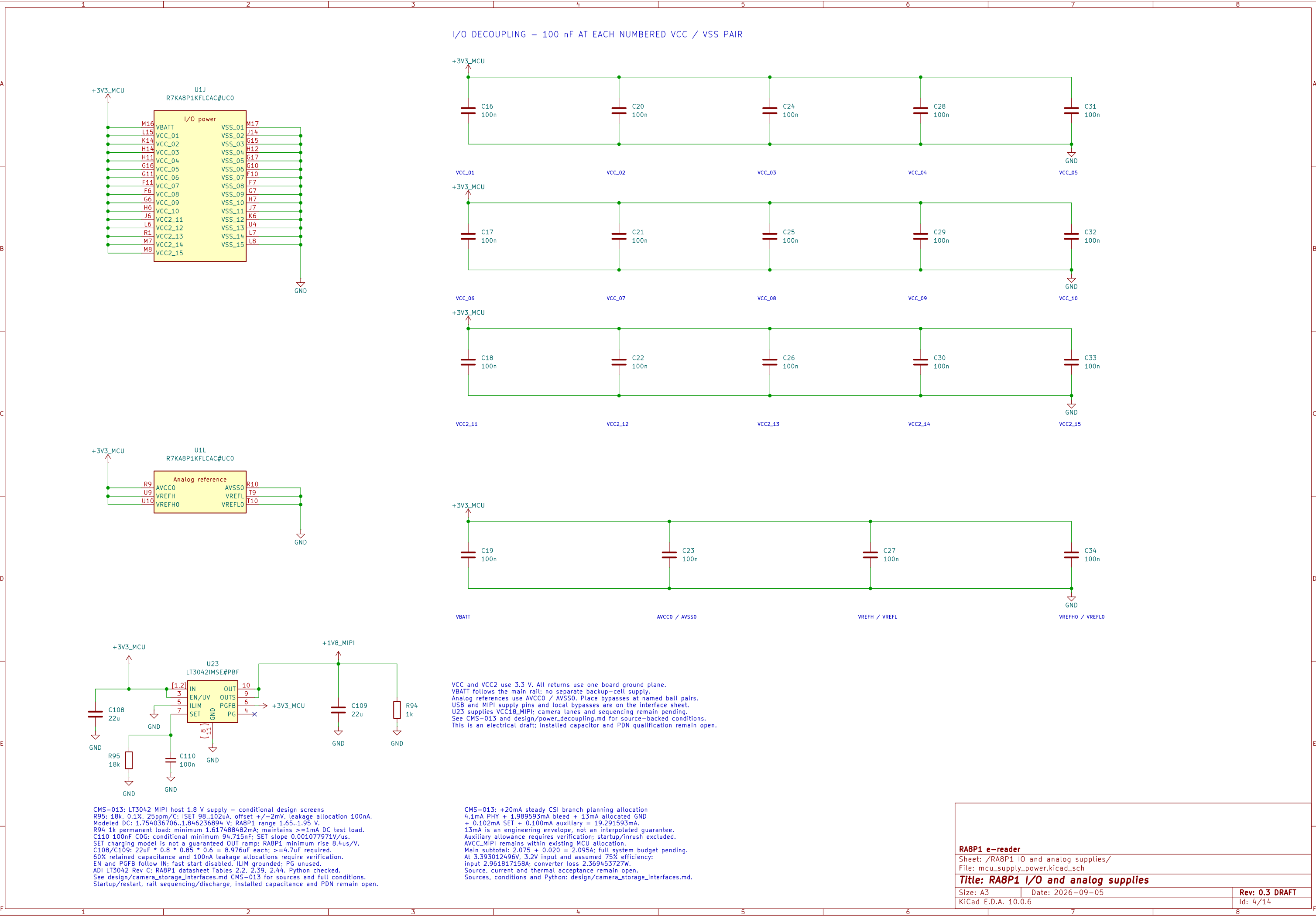
The schematic illustrates the I/O allocation for the RABP1 e-reader, organized by component and signal type. Key components include:

- Power Management:** USB-HS analog supply (FB1), +3V3_MCU, +3V3_USBHS_A, and various capacitors (C63, C40, C41, C42).
- Memory:** SDRAM modules (U1B-U1H) with associated address, data, and control signals.
- GPIOs:** Various pins (P0-P15, P2-P15, P4-P15, P6-P15, P8-P15, PA-PB, PC-PD) connected to different peripherals.
- Peripherals:** Radio (U1E), NOR flash (U1F), SDRAM controllers (U1G, U1H), and other interface chips (U1I, U1J, U1K, U1L, U1M, U1N, U1O, U1P, U1Q, U1R, U1S, U1T, U1U, U1V, U1W, U1X, U1Y, U1Z).

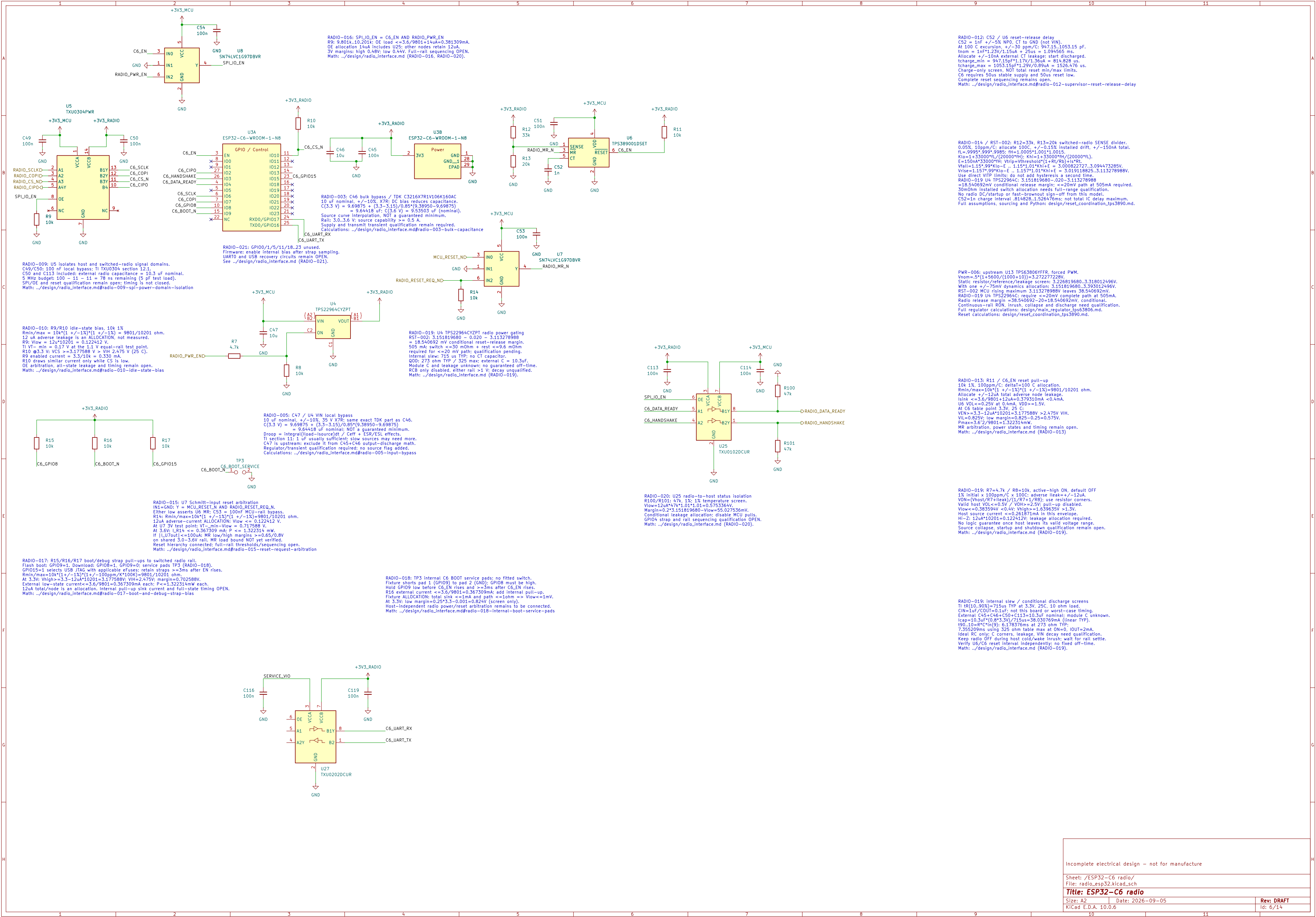
Key notes and calculations:

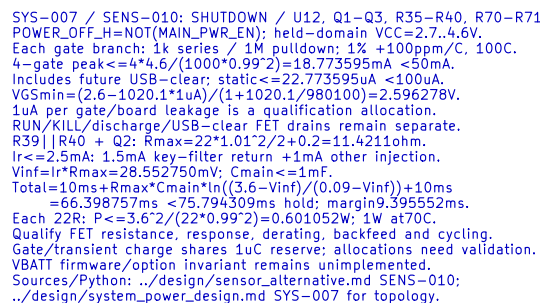
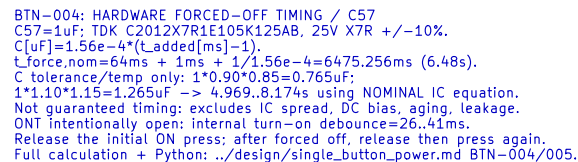
- USB-004:** USB-HS analog supply / FB1 + C63. FB1: 120 ohm at 100 MHz; DCR 0.05 ohm initial max. Screen with combined USB-HS current: I = 55.3 mA. Vdrop = IR = 2.765 mV; P = I^2 * R = 0.153 mW. After-test DCR 0.10 ohm: 5.53 mV / 0.306 mW. C63: 10 uF +/-10%, 35 V X7R; initial 9.11 uF. TDK nominal bias curve: 9.64 uF at 3.3 V. Not transient or all-corners qualification: verify rail ramp. Full calculations: ../design/usb_interface.md [USB-004]. ERC: #F1G03 declares the +3V3_MCU feed through FB1. This does not qualify the pending upstream SYS_AON source.
- CMS-016:** SCIO_SCK=P601; CIPO=P602; COPI=P603; CS_N=P604. READY=P706/BQ26; HANDSHAKE=P705/BQ19. P706=RESET_REQ_N; P707=PWR_EN. Host-domain signals. Timing, switched-domain isolation and reset arbitration remain open.
- USB-002:** USB full-speed supply / U1.U17. C40. Renesas QDG Table 1, p.6: 3.3 V with 100 nF bypass. C40 = 100*(1+/-0.10) = 90..110 nF initial: 50 V X7R. Nominal DC-bias estimate: 99.45 nF at 3.3 V (PWR-001). Place C40 close to VCC_USB with a short VSS_USB return. Full basis + Python check: design/usb_interface.md
- USB-003:** USB-HS supply / U1.H15. C41, C42. Renesas QDG Table 1 p.6: 10 nF ceramic + 47 uF electrolytic. C41 = 10*(1+/-0.10) = 9..11 nF initial: 50 V X7R. TDK nominal bias model: C41(3.3 V) = 10.084 nF (not guaranteed). C42 = 47*(1+/-0.20) = 37.6..56.4 uF initial: 6.3 V. Voltage screening: 3.6/6.3*100 = 57.14%; rating, not rail sign-off. C42 positive to 3V3, negative to GND; no reverse voltage. C41 close to H15; short returns to VSS1/VSS2_USBHS. Full math + Python / qualification: ../design/usb_interface.md
- PWR-001:** C39 AVCC_MIPI bypass, 100 nF (QDG Table 2, p.7). TDK C1608X7R1H104K080AA; also used at C6 and C16-C34. Initial C = 10



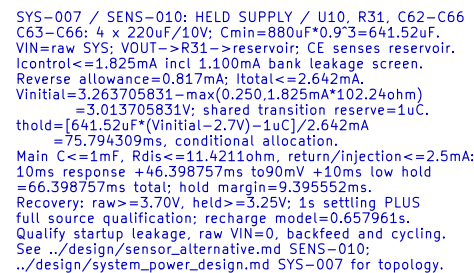




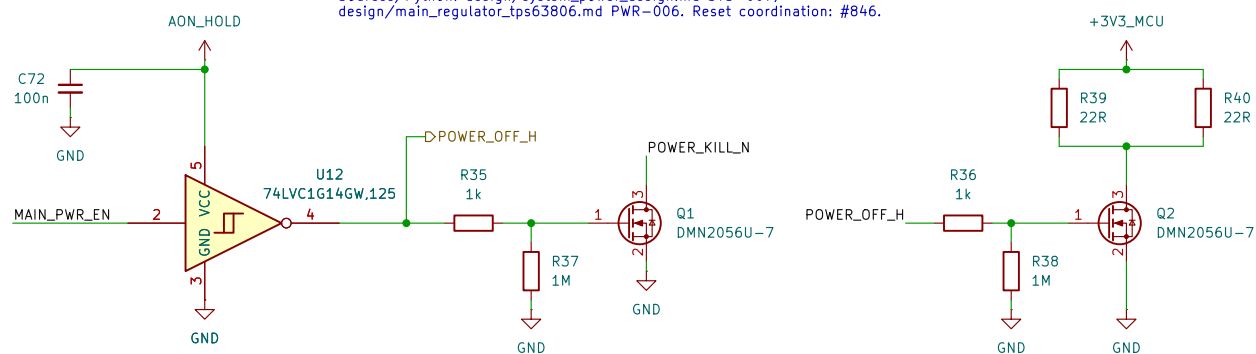


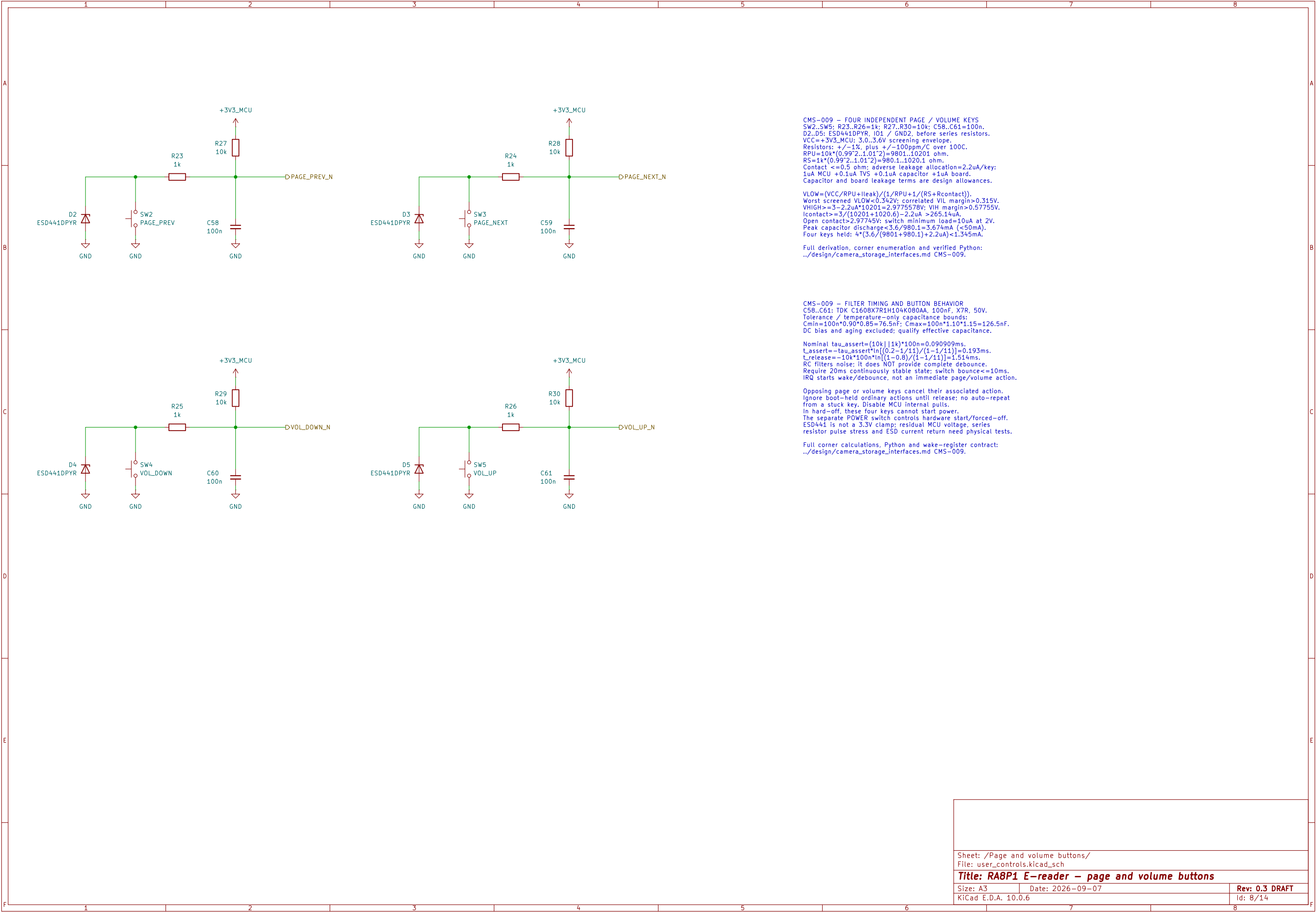


ERC supply path: U10 VOUT through R31.
PWR_FLAG declares that series-fed source only;
raw SYS_AON source remains unimplemented.



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SYS=007 / PWR=006: SOURCE SUPERVISION / U11, R32-R34, C67-C71
VtRpe=Vref*(1+R32/R33)+Isense*R32.
Nominal=0.405*(1+732k/100k)=3.369600V.
Vref +/-2%, Isense +/-25nA; each R +/-0.1%, 25ppm/C. 100C.
Corner trip=3.263705831...3.476597632V.
C68: tau=(732k||100k)*1n=87.980769ns (nominal).
C69-C71: 3x100n COG; initial/temp=284.445...316.827nF.
CT delay=300/(175+0.0005*1.714786k) nominal.
0.6*(284.445/175+0.0005)=0.974511s MODEL, not guarantee.
Require qualified reset hold >=90ns >650ms KILL blank +10ms.
R34=110k; ENhigh>=2.7-3.8uA*10k*1.01^2=2.273598V.
P0.0 sin(0.90sin(2*pi*100Hz))<1.0V, 0.0304V.
MAIN_PWR_EN_driver<=U14 buffer; held Q3 clamp; TP563806 EN.
Held U12 also drives KILL/discharge; USB-clear still needed.
Current EN network and rail calculations: PWR=006.
Sources/Python: design/system_power_design.m SYS=007;
design/main_regulator_tps63806.m PWR=006. Reset coordination: #846.
```





VLOW=(VCC/RP/Ileak)/(1/RPU+1/(RS+Rcontact)).
Worst stressed VLOW<0.342V; correlated Vih margin>0.315V.
VHIGH>=3-2.2uA*10201=2.9775578V; Vih margin>0.57755V.
Icontact>=3/(10201+1020.6)-2.2uA >265.14uA.
Open contact=2.97745V: switch minimum load=10uA at 2V.
Peak capacitor discharge<3.6/9801=3.674mA (<50mA).
Four keys held: 4*(3.6/9801+9801)+2.2uA<1.345mA.

CMS-009 – FILTER TIMING AND BUTTON BEHAVIOR
C58..C61: TDK C1608X7R1H104K080AA, 100nF, X7R, 50V.
Tolerance / temperature-only capacitance bounds:
 $C_{min}=100n*0.90*0.85=76.5nF$; $C_{max}=100n*1.10*1.15=126.5nF$
DC bias and aging excluded; qualify effective capacitance.

Nominal tau_assert=(10k|1k)*100n=0.090909ms.
 $\tau_{\text{assert}} = \tau_{\text{assert}} \cdot \ln[(0.2 - 1/1)/(1 - 1/1)] = 0.193\text{ms}.$
 $\tau_{\text{release}} = -10k * 100n \cdot \ln[(1 - 0.8)/(1 - 1/1)] = 1.54\text{ms}.$
 RC filters noise; it does NOT provide complete debounce.
 Require 20ms continuously stable state; switch bounce <= 10ms.
 IRQ starts wake/debounce, not an immediate page/volume action.

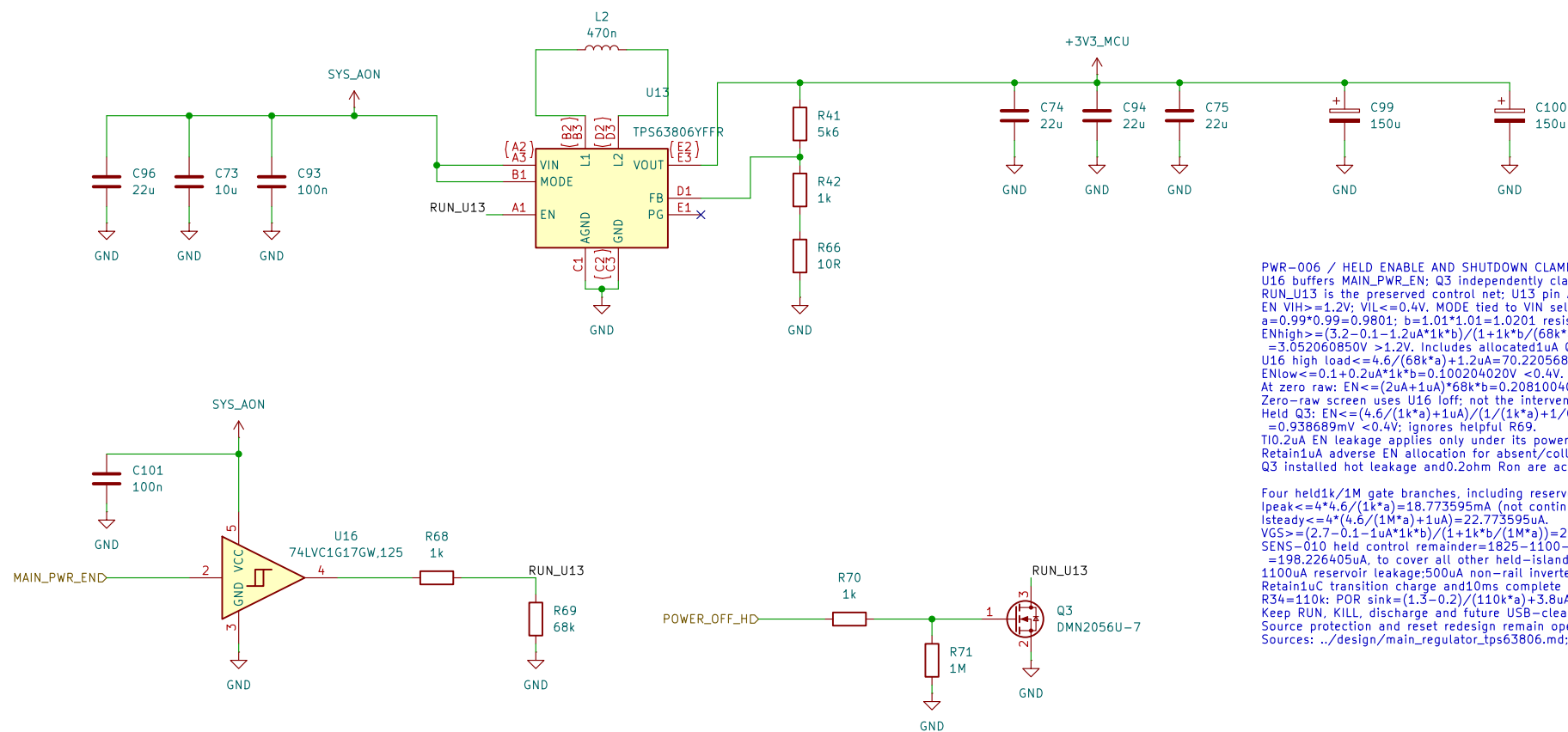
Opposing page volume keys cancel their associated action. Ignore boot-held ordinary actions until release; no auto-repeat from a stuck key. Disable MCU internal pulls.

In hard-off, these four keys cannot start power.

The separate POWER switch controls hardware start/forced-off. ESD441 is not a 3.3V clamp; residual MCU voltage, series resistor pulse stress and ESD current return need physical tests.

Full corner calculations, Python and wake-register contracts:
[../design/camera_storage_interfaces.md](#) CMS-009.

MAIN DIGITAL SUPPLY: TPS63806 / PWR-006
3.272277 V nominal; forced PWM; held shutdown clamp.
Divider and bypass calculations: design/main_regulator_tps63806.md
Reset coordination (#846), source protection and qualification remain open.
DEVELOPMENT SCHEMATIC – NOT FOR POWER-UP OR FABRICATION.



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PWR-006 / HLD ENABLE AND SHUTDOWN CLAMP
U16 buffers MAIN_PWD_EN; Q3 independently clamps EN during raw collapse.
RUN_U13 is the preserved control net; U13 pin A1 is now EN.
EN VIH>=1.2V; VIL<=0.4V. MODE tied to VIN selects forced PWM.
=0.99+0.99-0.9801; b=1.01*1.01+1.0201 (resistor factors)
= (3.2-0.1-1.2uA)*1k*b/(1+0.99)
= 3.052060850uA. Included allocationU13 off leakage.
U16 high load<=4.6/(68k*1)+1.2uA=70.220568uA <100uA.
ENlow=0.1+0.2uA*1k*b=0.100204020V <0.4V.
At zero raw: EN<=(2uA+1uA)*68k*b=0.208100400V.
Zero-raw screen uses U16 loff; not the intervening undefined region.
Held Q3: EN<=(4.6/(1k*1)+1uA)/(1/(1k*1)+1/0.2ohm)
=0.9386mV <0.4V; ignores held U16.
T10.2uA EN leakage applies only under its powered test conditions.
RetainU1A adverse EN allocation for absent/collapsing raw.
Q3 installed hot leakage and0.2ohm raw are acceptance allocations.

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Four held1k/1M gate branches, including reserved USB-clear:
Ipeak<=4*(1k^2/(1k^2a))=18.773595mA (not continuous ratings).
Isteady<=4*(1k^2/(1M^2a)+1UA)=22.773595mA.
VG55=(2.7-0.1*(1k+1k^2b))/(1k+1k^2b+1M^2a)=2.566278V.
SENS=held-control-remaining=862.5mA-200-500=22.773595
=198.226450uA, to cover all other held-island loads.
1100uA reservoir leakage:500uA non-rail inverter-input allocation.
RetainLTC transition charge and10ms complete response allocation.
R34=110k: POR sink=(1.3-0.2)/(110k^2a)+3.8uA=14.003041uA.
KILL=KILL: POR sink=1.3/(1k^2a)+3.8uA=14.003041uA, separate.
Source protection and reset redesign remain open; no power-up approval.
Source.../design/main_regulator_tps63806.m: SENS=0101 revision.

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PWR-Vom / FEEDBACK AND LOCAL PASSIVES
 $V_{nom} = 5.00 \times (1 + 5600 / (1000 + 10)) = 3.27227228 \text{ V}$
R41: 0.02%, 5ppm/C; R42: 0.01%, 2ppm/C; R66: 1%, 200ppm/C.
Installed boards: initial "TCR1000" aging*assembly, then ohms.
 $R41 = 5600 \times (1 + (-0.0002)) \times (1 + (-0.0005)) \times (1 + (-0.001))$
 $\times (1 + (-0.0005)) \div (-0.02) = 5587.6629237, 5612.349243 \text{ ohm}$
 $R42 = 1000 \times (1 + (-0.0001)) \times (1 + (-0.0002)) \times (1 + (-0.001))$
 $\times (1 + (-0.0005)) \div (-0.01) = 998.190970, 1001.810970 \text{ ohm}$
 $R66 = 10 \times (1 + (-0.01)) \times (1 + (-0.02)) \times (1 + (-0.01)) \div (-0.05)$
 $= 9.954980, 10.455020 \text{ ohm}$ once combined aging/assembly 1%.
 $V_{in} = 5.00 \times (1 + 14160 / (R41 \times R42 \times R66)) = 5.00 \times (1 + 14160 / 1 \times 1 \times 1)$
 $V_{in} = 5.005 \times (1 + 14160 / (R42 \times R66 \times 1)) = 5.005 \times (1 + 14160 / (1 \times 1 \times 1))$
 $V_{static} = 3.226819680, 3.318012496 \text{ V (FPWM only)}$
One combined + / -75mV regulation/ripple/transient/routing allowance:
 $V_{complete} = 3.151819680, 3.393012496 \text{ V}$: qualification required.
SDRAM static VOH margin = $2.4 - 0.73 = 3.93012496 \text{ V} = 24.891253 \text{ mV}$.
This is not measured S/I/noise headroom or reset qualification.

12: 470nH +/-20%; catalog spread alone is insufficient.
Screen incoming0.412..0.518uH; allocated operating+/-10%
= +/-3708..0.5698uH versus T137..0.57uH. Quality hot/current.
C74/C75/C94: *322uF*0.8*0.85^0.6=6.2928uF > T121uF
C96: 22uF*0.8*0.85^0.6=8.976uF > T14uF input minimum.
The capacitor combined resistance is not guaranteed.
Samsung typical output bias retention at3.393012496V-90.221554%.
Input bias, aging, mounted impedance and startup need verification.
Full assumptions, sources and executable Python: click this note.

PWR-006 / LDA, STORAGE AND SHUTDOWN
Allocation: 1.80A operating; 2.25A cold start screen, not measured load.
Pwr: 3.393012496e+1, 8=6.107424332
Thermal resistance: θ_{JA} allocated 75°C/W; θ_{JA} Pin=84.95329991W; $\ln$ pin 3.2V=2.544759372A.
Losses: 2.5807876W; θ_{JA} = an adiabatic continuous thermal result.
AT90C efficiency: loss=0.678602499W.
TJ50°C=70°C+78.8°C/0.678602499W=123.474C.
2.54V theta-JA is not the final PCB thermal resistance.
TJ=84.95329991W; θ_{JA} Pin=84.95329991W; $\ln$ pin 3.2V=2.544759372A.
Converter limit, source limit, thermal and thermal envelope need validation.

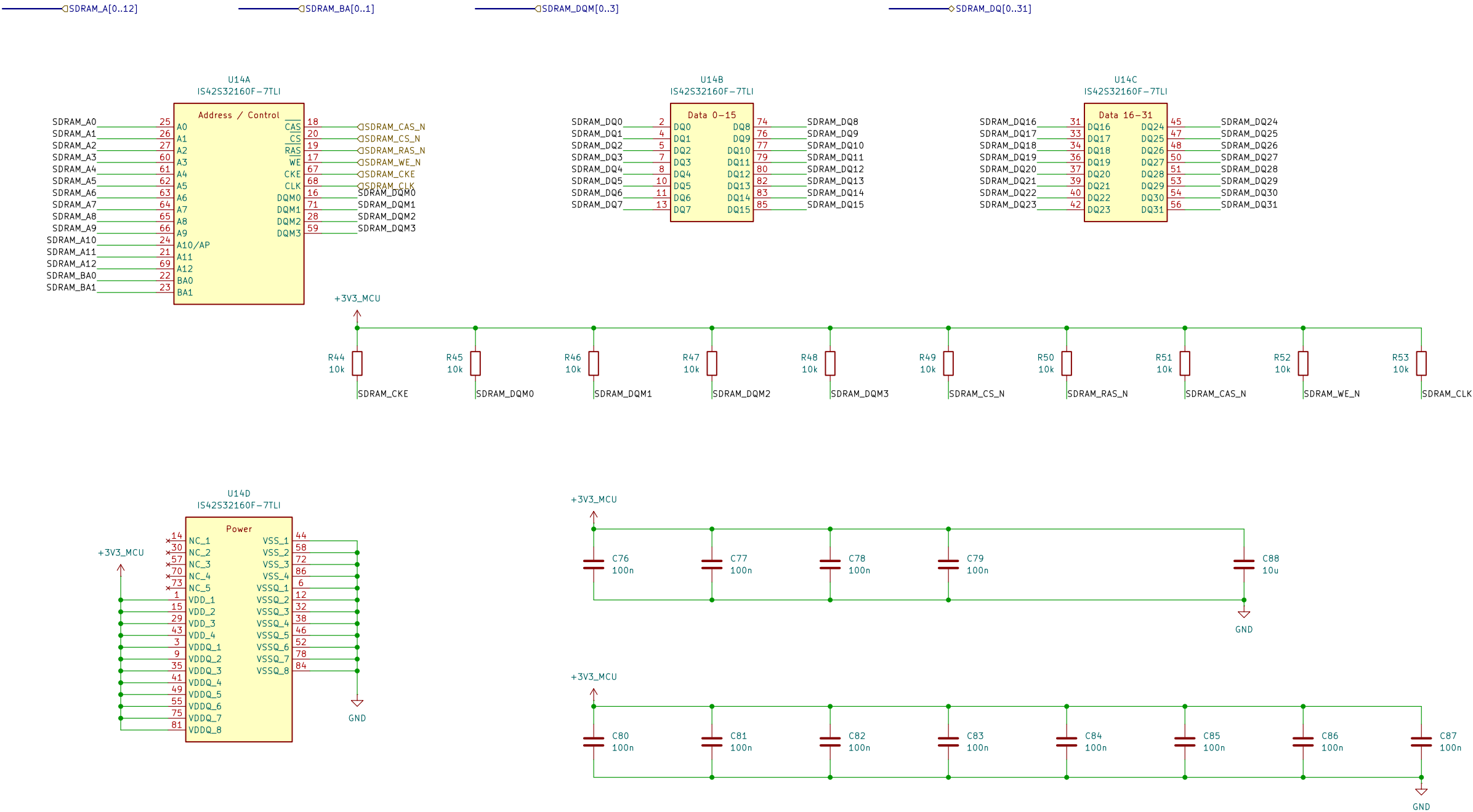
PWR=006 storage checkpoint (nominal, not effective minimum):
Direct main469.21uF + C43 through FB1 10uF + key filters0.4uF
=479.61uF cold; radio-on add±0.2uF =>489.81uF.
C99/C100 remain±150uF each. C94 adds22uF to the output.
Old bootstrap/private VCC/VyF capacitors were not main-rail storage.
Whole-main ceiling remains1mF; reconcile later peripheral additions.
Cout lower acceptance and distributed PDN/startup are separate checks.

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SENS-010 discharge revision: Rmax1x1.42110hm: return/injection<=2.5mA.
Includes1.5mA key return +1mA other backfeed: Vinf=0.028552750V.
t=1mF1x1.42110hm*ln((3.6-Vinf)/(0.09-Vinf))+10ms+10ms
=66.3987575ms total, below475.794309ms screen.
Includes1.0ms delay and ramp/discharge to 10ms below90mV.
Held budget:1.825mA control +0.817mA reverse;1uC transition reserve.
198.226405uA control remainder must cover other held-island loads.
SYS_AON source protection, loaded startup and fast faults remain open.
RSN_002 / RADIO-019: release margin remains conditional on rail/path bounds.
No more power or current margin for control and coordination section are complete.
Full assumptions: ../design/main_regulator_tps63806.mt
shutdown revision/Python: ../design/sensor_alternative.md SENS-010.

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64 MiB SDRAM | ISSI IS42S32160F-7TLI
Signal and reset—default circuits | CMS-010.
Startup, SI/timing and hardware qualification remain open.



CMS-010 | SDRAM power and bypass
VDD and VDDQ share +3V3_MCU (3.0 to 3.6 V).
C76-C79: 4 x 100 nF for the four VDD pins.
C80-C87: 8 x 100 nF for the eight VDDQ pins.
C88: 10 uF local bulk; Cnom = 12 x 0.1 + 10 = 11.2 uF.
Included in the PWR-003 1 mF whole-rail limit.
Effective C, placement and PDN impedance require qualification.

CMS-010 | R44-R53: 10k reset defaults
Rail 3.0..3.6V: +/-1%, 100ppm/C. |dT| <=100C.
Rmin=10k*0.99*0.99=9801 ohm.
Rmax=10k*1.01*1.01=10201 ohm.
Ioff=5uA SDRAM+1uA MCU+1uA board=7uA.
VHIGHmin=3.0-7uA*10201=2.928593V (>2.0V VIH).
Board leakage and temperature screen require qualification.

CMS-010 | Pull load and power states
Sink leakage: 5uA SDRAM+1uA board=6uA.
Isink=3.6/9801+6uA=0.373309mA (<0.374mA).
Pmax=3.6^2/9801=1.322314mW per pull.
All ten LOW: 10*3.6/9801=3.673095mA rail:
10*3.6^2/9801=13.223140mW resistor heat.
Reserve 4mA within the 250mA SDRAM allocation.
CKE/DQM and CS_N default HIGH; CLK HIGH is not a clock.
Switched-rail pulls; self-refresh actively holds CKE LOW.
Startup, dynamic I/O, SI and shutdown qualification OPEN.

Sheet: /64 MiB SDRAM/
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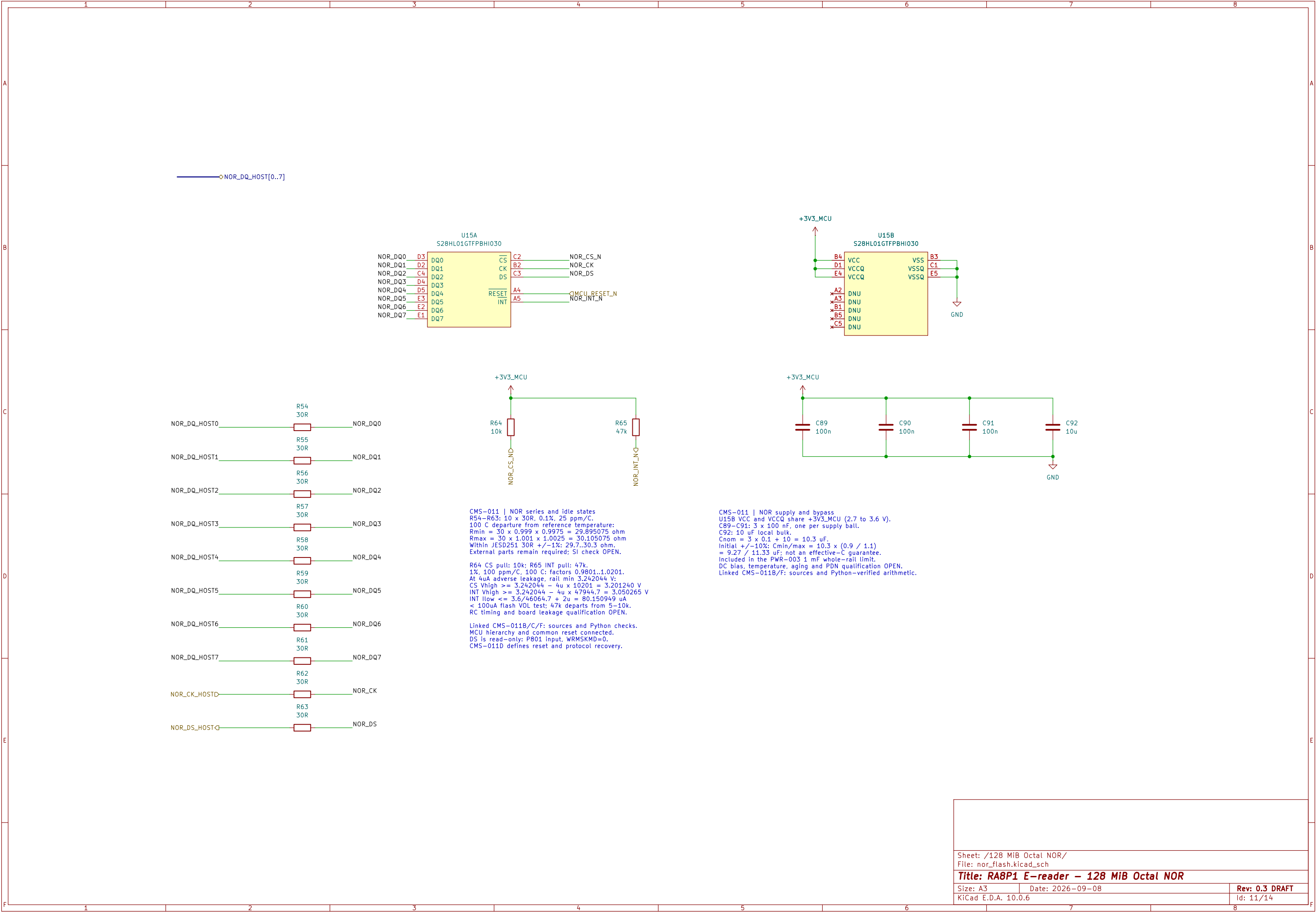
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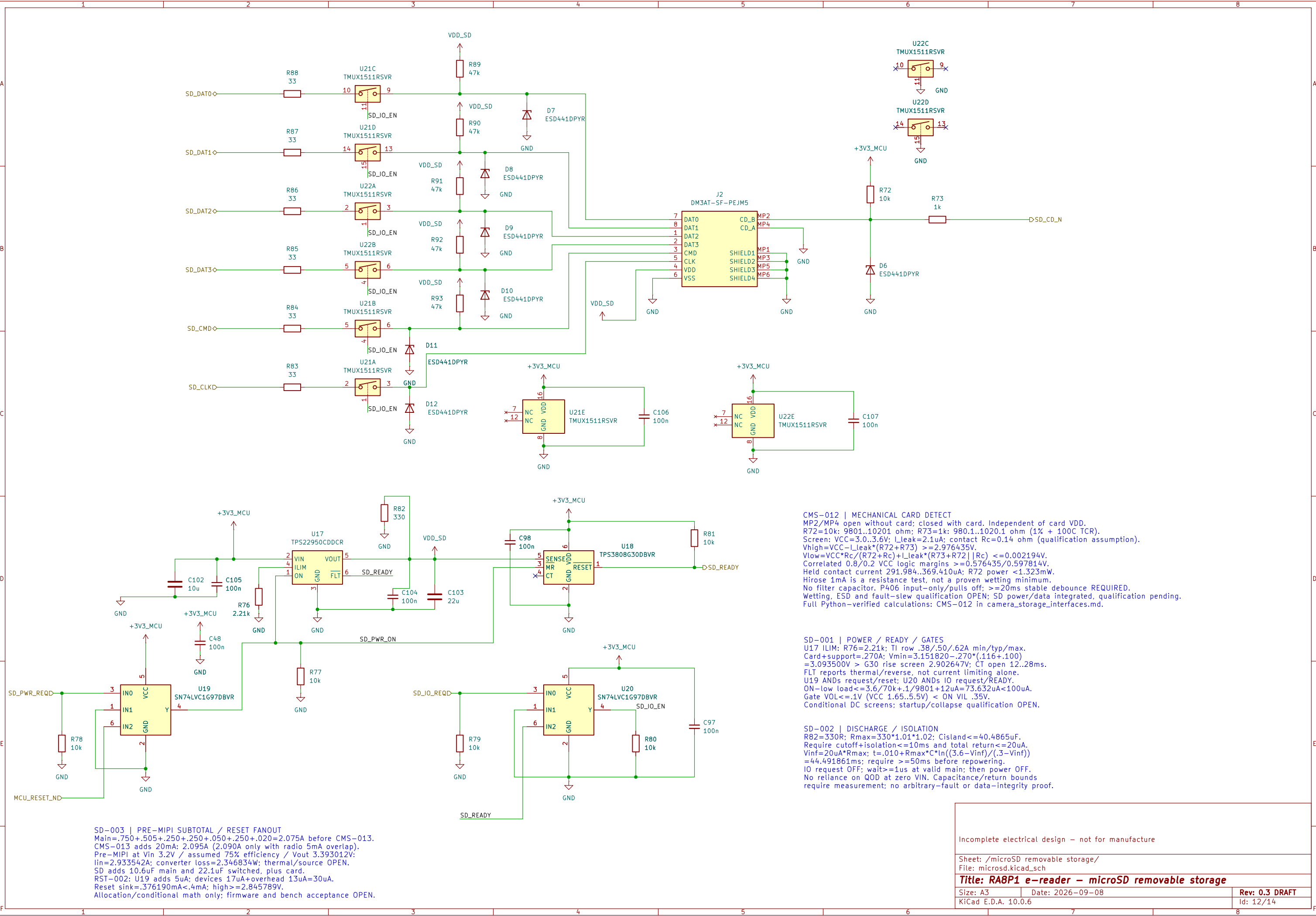
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Rev: 0.3 DRAFT

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SD-003 | PRE-MIPI SUBTOTAL / RESET FANOUT
Main=.750+.505+.250+.250+.050+.250+.020=2.075A before CMS-013.
CMS-013 adds 20mA: 2.095A (2.090A only with radio 5mA overlap).
Pre-MIPI at Vin 3.2V / assumed 75% efficiency / Vout 3.393012V:
Iin=2.933542A; converter loss=2.346834W; thermal/source OPEN.
SD adds 10.6uF main and 22.1uF switched, plus card.
RST-002: U19 adds 5uA; devices 17uA+overhead 13uA=30uA.
Reset sink=.376190mA<.4mA; high>=2.845789V.
Allocation/conditional math only; firmware and bench acceptance OPEN.

CMS-012 | MECHANICAL CARD DETECT
MP2/MP4 open without card; closed with card. Independent of card VDD.
R72=10k; 9801..10201 ohm; R73=1k; 9801..1020.1 ohm (1% + 100C TCR).
Screen: VCC=3.0..3.6V; ILeak=2.1uA; contact Rc=0.14 ohm (qualification assumption).
Vhigh=VCC-ILeak*(R72+R73) >=2.976435V.
Vlow=VCC*Rc/(R72+Rc)+ILeak*(R73+R72) | Rc) <=0.002194V.
Correlated 0.8/0.2 VCC logic margins >=0.576435/0.597814V.
Held contact current 291.984..369.410uA; R72 power <1.323mW.
Hiros 1mA is a resistance test, not a proven wetting minimum.
No filter capacitor. P406 input-only/pulls off; >=20ms stable debounce REQUIRED.
Wetting, ESD and fault-slew qualification OPEN; SD power/data integrated, qualification pending.
Full Python-verified calculations: CMS-012 in camera_storage_interfaces.md.

SD-001 | POWER / READY / GATES
U17 ILIM: R76=2.21k; TI row .38/.50/.62A min/typ/max.
Card+support=.270A; Vmin=3.151820-.270*(.116+.100)
=3.093500V > G30 rise screen 2.902647V; CT open 12..28ms.
FLT reports thermal/reverse, not current limiting alone.
U19 ANDs request/reset; U20 ANDs IO request/READY.
ON-low load<=3.6/70k+.1/9801+12uA=73.632uA<100uA.
Gate VOL<=.1V (VCC 1.65..5.5V) < ON VIL .35V.
Conditional DC screens; startup/collapse qualification OPEN.

SD-002 | DISCHARGE / ISOLATION
R82=330R; Rmax=330*1.01*1.02; Cisland<=40.4865uF.
Require cutoff/isolation<=10ms and total return<=20uA.
Vinf=20uA*Rmax; t=.010+Rmax*C*ln((3.6-Vinf)/(.3-Vinf))
=44.491861ms; require >=50ms before repowering.
IO request OFF; wait>=1us at valid main; then power OFF.
No reliance on QOD at zero VIN. Capacitance/return bounds
require measurement; no arbitrary-fault or data-integrity proof.

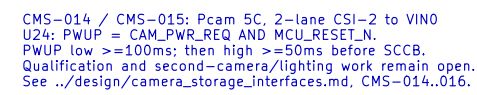
Incomplete electrical design – not for manufacture

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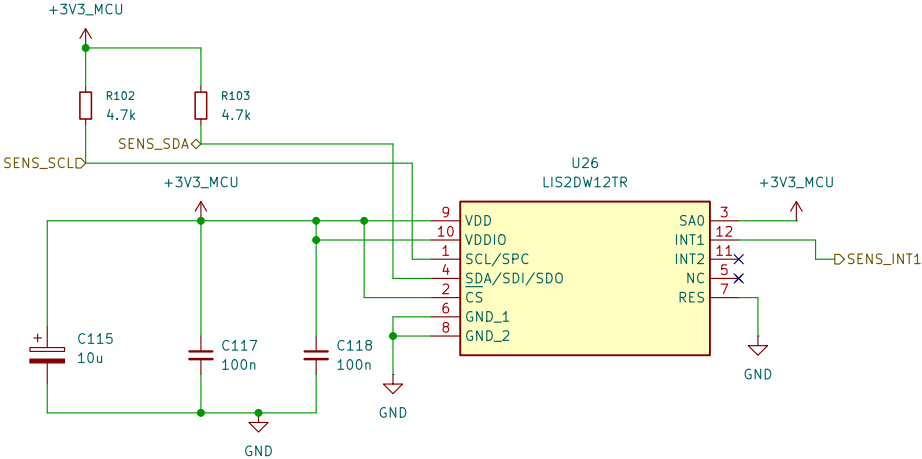
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ORIENTATION SENSOR – WIP
U26: LIS2DW12TR; ST DS11811 Rev 9 pin map.
VDD pin 9 / VDDIO pin 10: +3V3_MCU; CS / SA0 high for I2C address 0x19.
C117/C118: local 100n bypass; GND pins 6/8 and RES pin 7 grounded.
C115: 10u/16V aluminum EEE-FM1C100R; place near VDD pin 9 with C117.
R102/R103: 4.7k SCL/SDA pull-ups; obsolete ADXL VREG network removed.
Host connected: SCL P410/K13; SDA P409/R16; INT1 P306/D11.
Wait 20ms after supply valid. Power-cycle below 100mV for >=10ms.
Pending: sourcing fields, pin modes, timing/power/passive qualification.
See design/sensor_alternative.md; circuit qualification remains open.



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